MIL-B-80247B
29 May 1987
SUPERSEDING
MIL-B-80247A
23 January 1979

MILITARY SPECIFICATION

BENDING MACHINES, SHEET AND PLATE, CYLINDRICAL RESILIENT ROLLER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers resilient roll bending machines for bending sheet metal and plate into cylindrical shapes.

1.2 Classification. The bending machines shall be of the following types and sizes listed in tables I, II, and III. The types and sizes, to be furnished shall be as specified (see 6.2.1).

Type I - Manually Operated.

Type II - Pneumatically Operated.

Type III - Hydraulically Operated.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441
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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.12 - Safety Requirements for Construction, Care, and Use of Roll-Forming and Roll-Bending Machines.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

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JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

JIC P-1 - Pneumatic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the Publication Sales Division, National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The bending machine shall be new and one of the manufacturer's current models capable of forming sheet metal and plate in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

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3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The bending machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The bending machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the bending machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR, Chapter XVII, Part 1910 and ANSI B11.12 that are applicable to the machine itself. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. The bending machine shall be so designed and constructed that, under the operation, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation .

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3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (6.2.1), an automatic centralized lubrication system shall be provided.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The bending machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

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3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The bending machine shall consist of not less than the following components.

3.4.1 Frame. The frame shall be constructed with the required strength and rigidity to support and maintain alignment of all components of the machine while operating at its maximum capacity. Components mounted within the frame shall be accessible for maintenance. Anchor bolt holes shall be provided to secure the machine to the floor.

3.4.2 Rolls. The machine shall be equipped with one steel roll and one resilient roll. The steel roll shall have the hardness, surface finish, and toughness to resist wear and prevent marring of the workpiece. The resilient roll shall be of a urethane elastomer having chemical and mechanical properties suitable for the intended use.

3.4.2.1 Slip-on tubes. When required steel slip-on tubes for forming diameters larger than the maximum capacity diameter of the steel roll shall be furnished. The slip-on tubes shall be of the length and diameter specified (see 6.2.1). Instructions for the development and machining of slip-on tubes shall be furnished with each machine.

3.4.3 Roll position indicator. A mechanical roll position indicator, readable from the operating position, shall be furnished to enable the operator to accurately indentify and set the position of the resilient roll for duplicate work.

3.4.4 Roll adjustment control. The top roll shall be in a fixed position. The resilient roll shall be vertically adjustable based on the material, slip-on tube thickness, and the required forming pressure. The resilient roll on the type I machines shall be manually adjustable. Unless otherwise specified (see 6.2.1), the resilient roll on the type II machines shall be pneumatically adjustable and the resilient roll on type III machines shall be hydraulically adjustable. Roll adjustment controls shall be located convenient to the operator at his normal work station.

3.4.4.1 Roll pressure lockup. The resilient roll shall have a locking device to apply and relieve the forming pressure between the rolls. When lowered the locking device shall relieve the roll pressure to facilitate workpiece removal without changing roll adjustment. When the locking device is reengaged the resilient roll shall return to the proper position to form a duplicate workpiece.

3.4.5 Roll drive system. The lower roll shall be driven either by an electric motor with reduction gear drive or a hydraulic drive system capable of providing smooth no-hesitant roll speeds conforming to the requirements of tables I, II, or III as applicable to the machine required. The roll drive system shall have forward, reverse, and instant stop with controls located conveniently to the operator at his normal work station.

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3.4.6 Upper roll support. The upper roll shall be secured in the operating position by a stationary bearing support on one end and a retractable roll support on the free end. The roll support, when lowered, shall allow removal of the workpiece. The roll support for the type II machine shall be powered pneumatically and the type III machine hydraulically. The controls for the support shall prevent the rolls from making contact unless the support is in the operating position. A locking device shall secure the top roll and roll support in position.

3.4.7 Workpiece guide. The machine shall be equipped with a guide of the manufacturer's standard commerical design normally furnished on the size machine supplied. The guide shall be used to square the side of the work.

3.4.8 Pneumatic system. The pneumatic system on the type II machines shall be operated from an outside air source and shall consist of at least a regulator, filter, control valve, and the necessary cylinders, hoses, and piping. When specified (see 6.2.1), the pneumatic system shall conform to JIC P-1.

TABLE I. Type I sizes and capacities.

Size	Effective working width (inches)	Diameter of speed roll (inches)	Rolling speed (feet per minute)	Material thickness gauge	Drive motor horsepower
18 x 3 1/2	18	3 1/2	16	16	1
24 x 3 1/2	24	3 1/2	16	18	1

NOTE: All requirements shall be not less than those shown.

TABLE II. Type II sizes and capacities.

Size	Effective working width (inches)	Diameter of speed roll (inches)	Rolling speed (feet per minute)	Material thickness (gauge or inches)	Drive motor horsepower
18 x 3 1/2	18	3 1/2	20	14 ga.	1 1/2
18 x 5	18	5	20	3/16 in.	3
24 x 4	24	4	20	12 ga.	1 1/2
24 x 5	24	5	20	10 ga.	3
30 x 5	30	5	20	14 ga.	1 1/2

NOTE: All requirements shall be not less than those shown.

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TABLE III. Type III sizes and capacities.

Size	Effective working width (inches)	Diameter of speed roll (inches)	Rolling speed (feet per minute)	Material thickness (gauge or inches)	Drive motor horsepower
18 x 6	18	6	18	3/16 in.	5
24 x 3	24	3 1/2	20	12 ga.	4
24 x 6	24	6	18	10 ga.	5
24 x 7	24	7	18	3/16 in.	5
30 x 6	30	6	19	11 ga.	3
30 x 7	30	7	18	10 ga.	5
30 x 9 1/2	30	9 1/2	15	1/4 in.	7 1/2
36 x 6	36	6	19	12 ga.	3
36 x 8	36	8	12	10 ga.	5
36 x 9 1/2	36	9 1/2	15	1/4 in.	7 1/2
48 x 7	48	7	20	12 ga.	3
48 x 9 1/2	48	9 1/2	15	3/16 in.	7 1/2
48 x 12	48	12	12	3/16 in.	7 1/2
60 x 9 1/2	60	9 1/2	15	11 ga.	7 1/2
60 x 12	60	12	16	9 ga.	12 1/2
72 x 9 1/2	72	9 1/2	15	11 ga.	7 1/2
72 x 12	72	12	16	9 ga.	12 1/2
89 x 9 1/2	89	9 1/2	15	14 ga.	7 1/2

NOTE: All requirements shall be not less than those shown.

3.4.9 Hydraulic system. The hydraulic system on the type III machines shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. When specified (see 6.2.1), the hydraulic system shall conform to JIC H-1.

3.4.10 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. Unless otherwise specified (see 6.2.1), all AC motors and and controls shall be designed for operation on 60-Hz electrical power.

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3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.10.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.10.3 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I, II, or III for the machine size specified.

3.6 Performance. The machine shall be capable of rolling sheet or plate of the thickness and width specified in tables I, II, or III and having a carbon content of not more that 0.10 percent into a cylinder with a minimum inside diameter of not more than one inch larger than the steel roll. The maximum diameter shall be limited only by work space above the machine. When tested in accordance with 4.7.2.1, the formed cylinder shall have no visible damage to the surface contacted by the resilient roll. The contour of the cylinder shall deviate not more than 0.5 percent excluding the leading and trailing edge flats. The edge flats on the formed workpiece shall be not greater than four times material thickness. The leading and trailing edges of the formed workpiece shall be parallel within 0.125 inch per foot of rolled cylinder length.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not re-

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lieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements sepcified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. Each bending machine shall form a workpiece of the capacity width and thickness shown in tables I, II, or III into a cylinder having an inside diameter not more than 1 inch larger than the steel roll. The cylinder shall be formed in one pass. The contour, parallelism of edges, and end flats shall conform to the requirements of 3.6

4.7.2.1 Surface damage test. The surface damage test shall be performed using a polished, annealed aluminum sheet not less than the machine's effective working width with a thickness not less than 20 gauge. The workpiece shall be rolled to form a cylinder of the minimum diameter for the machine tested. There shall be no visible evidence of scratches or abrasions of the finished surface due to contact with the resilient roll.

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4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended for forming cylindrical shapes from polished or painted workpieces without damage to the finished surface.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US or SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.4).

h. Automatic centralized lubrication system, if required (see 3.2.8).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

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k. Length and diameter of slip-on tubes, if required (see 3.4.2.1).

l. Resilient roll adjustment, if different (see 3.4.4).

m. Pneumatic system, in conformance with JIC P-1, if required (see 3.4.8).

n. Hydraulic system, in conformance with JIC H-1, if required (3.4.9).

o. Electrical system and input power, if different (see 3.4.10).

p. Motor and control frequency, if different (see 3.4.10).

q. Motor enclosure, if different (see 3.4.10.1).

r. Size and capacity, if different (see 3.5).

s. Additional performance features, if required (see 3.6.1).

t. Standard equipment, if different (see 3.7).

u. Specify and fully describe optional equipment, if required (see 3.8).

v. Specify applicable repair parts, if required (see 3.9).

w. Hour meter, if different (see 3.11).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (see 4.5).

z. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

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MIL-B-80247A	MIL-B-80247B
Type I	Type I Size 24 x 3 1/2 (new size)
Type II Size 18 x 4	Type II Deleted
Type III	Type III
Size 30 x 8	Size 24 x 3 1/2 (new size) Deleted
Size 36 x 7	Size 30 x 9 1/2 (new size) Deleted
Size 48 x 8	Size 36 x 9 1/2 (new size) Deleted
Size 60 x 8	Size 48 x 9 1/2 (new size) Deleted
Size 72 x 8	Size 60 x 9 1/2 (new size) Deleted
	Size 72 x 12 (new size) Size 89 x 9 1/2 (new size)

6.5 Subject term (key word) listing.

Bending
Cylindrical
Forming
Machine
Metal, sheet, bending, drawing, working
Resilient
Roll, forming

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Air Force-84,99
DLA-GS

User activity:
Navy - MC

Preparing activity:
DLA - IP

(Project 3441-0123)

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